

Program Question_4

```
Implicit none
Real::a=-0.4,b=0.6,h,fa,fb,fc,Trapezoidal_cal,simpsons_cal,simpsons_third_eight,Exact_value
Real::abs_Error1, Relative_Error1
Real:: abs_Error2, Relative_Error2
Real::abs_Error3, Relative_Error3
!Compute exact value

Exact_value=f_exact(a)-f_exact(b)
!Compute stepsize h

h=b-a
!Compute functional value f(a) and f(b)
fa=f(a)
fb=f(b)
fc=f(a+h)
!Using Trapezoidal Rule
Trapezoidal_cal=(h/2.0)*(fa+fb)
abs_Error1=Abs(Exact_value-Trapezoidal_cal)
Relative_Error1=Abs((Exact_value-Trapezoidal_cal)/Exact_value)
!Using Simpsons 1/3 rule
simpsons_cal=(h/3.0)*(fa+fc+fb)
abs_Error1=Abs(Exact_value-simpsons_cal)
Relative_Error1=Abs((Exact_value-simpsons_cal)/Exact_value)

!Using Simpsons 3/8 rule
simpsons_third_eight=(3.0*h/8.0)*(f(a)+3*f(a+h)+3*f(a+2*h)+f(b))
abs_Error3=Abs(Exact_value-simpsons_third_eight)
Relative_Error3=(Abs(Exact_value-simpsons_third_eight)/Exact_value)

Write(*,*)"Method      |Approximation      |Absolute_Error      |Relative_Error      |Exact_Value"
Write(*,*)"-----"
Write(*,*)"Trapezoidal ", Trapezoidal_cal,abs_Error1, Relative_Error1,Exact_value
Write(*,*)"Simpsons 1/3", simpsons_cal,abs_Error2, Relative_Error2,Exact_value
Write(*,*)"Simpsons 3/8", simpsons_third_eight,abs_Error3, Relative_Error3,Exact_value

contains
Real Function f(x)
Real,Intent(In)::x
f=- (1/((x-1)*(2+sqrt(2.0)-2*x)))
End Function

Real Function f_exact(x)
Real,Intent(In)::x
f_exact=((-1/sqrt(2.0))*Log(Abs((x-1)/(sqrt(2.0)-2*(x-1)))))
End Function
```

End Program